Health Risk Intelligence Platform

Comprehensive Technical Specification & Side-by-Side Audit Report

Clinical Focus: *Type 2 Diabetes Stratification, Explainability & Counterfactual Interventions* | ***Version:*** *2.1-Production* | ***Status:*** *Validated*

1. Executive Summary & Clinical Mission

The Health Risk Intelligence Platform is an end-to-end, clinical-grade medical decision support system designed to transcend basic academic modeling. Traditional health risk notebooks print uncalibrated accuracy scores on unvalidated data splits. In contrast, this platform provides **calibrated empirical risk probabilities**, **SHAP game-theory biomarker attributions**, an interactive **Counterfactual 'What-If' Simulation Engine**, and **evidence-based ADA clinical care plans**.

2. Side-by-Side Comparative Audit: Baseline vs. Enhanced Platform

Evaluation Dimension	Original Baseline Version	Enhanced Production Platform (v2.1)
Model Architectures	Single script with 3 models; no modern histogram tree algorithms.	**5-Model Tournament: Logistic Regression, Random Forest, XGBoost, LightGBM (HistGradient), and Calibrated models.**
Evaluation Matrix	AUPRC and AUROC only.	**Full Matrix: AUPRC, AUROC, Brier Score, F1, Accuracy, Precision, Recall/Sensitivity, Specificity.**
User Authentication	Hardcoded dummy string ('demo_user'). No security.	**JWT Bearer Auth with bcrypt salted hashing (12 rounds) & role permissions (Clinician, Admin).**
Decision Support	Static risk score only; zero actionable next steps.	**ADA Clinical Action Plan: Prioritized diet, exercise, and lab orders mapped to SHAP drivers.**
Counterfactual Reasoning	None. Patient cannot test lifestyle modifications.	**What-If Simulator: Interactive sliders calculate exact risk reduction in real time.**
EHR Integration	Standard browser window print.	**Hospital-Grade EHR SOAP Note Generator with diagnostic ICD-10 billing codes (Z13.1, R73.03, E66.01).**
Cardio-Metabolic Screening	Diabetes probability only.	**ATP III / IDF Cardio-Metabolic Syndrome Index: Checks 5 co-occurring cardiovascular risk factors.**
Body Composition Usability	Forced user to know exact decimal BMI number.	**Integrated BMI Calculator: Converts Height & Weight in both Metric (cm/kg) and Imperial (ft/in/lbs).**

3. Multi-Algorithm Tournament Performance (Held-Out Test Set N=1,554 from 7,768 Multi-Center Cohort)

Algorithm Architecture	Family	Test AUPRC	Test AUROC	Brier Score	Sensitivity	Specificity	Accuracy
Logistic Regression (Winner) ■	**Generalized Linear**	**0.7734**	**0.8513**	**0.1461**	**68.9%**	**84.0%**	**79.4%**
Calibrated Logistic Reg	Isotonic Post-Hoc	0.7713	0.8512	0.1304	53.9%	94.2%	82.0%
LightGBM (HistGradient)	Histogram-Binned Trees	0.7656	0.8460	0.1451	67.4%	84.1%	79.1%
Random Forest (300 Trees)	Bagging Ensemble	0.7653	0.8478	0.1455	65.2%	86.6%	80.2%
XGBoost (Depth 4)	Gradient Boosted Trees	0.7594	0.8411	0.1460	67.6%	85.3%	79.9%

4. Clinical Data Engineering & Explainable AI (SHAP)

Zero-as-Missing Sanitization: In clinical datasets, physiologically impossible zeros (e.g. Glucose = 0 or BP = 0) represent uncollected laboratory tests rather than true zero values. Our pipeline converts zeros to NaN and performs median imputation exclusively on training folds to prevent data leakage.
Clinically-Grounded Features: Synthesizes the Metabolic Synergy Index (Glucose * BMI) / 100, WHO BMI buckets, and ADA glycemic classifications.
SHAP Cooperative Game Theory: Every prediction decomposes into additive marginal contributions $f(x) = \phi_0 + \sum \phi_i$ against a 100-sample reference background distribution, enabling clinicians to audit exact biomarker attributions.

5. Microservice Architecture & API Directory

Endpoint	HTTP Method	Clinical Functionality
/predict	POST	Calibrated risk probability, population percentile, SHAP drivers, and ADA care plan.
/predict/simulate	POST	Counterfactual What-If simulation: calculates exact risk reduction when altering biomarkers.
/predict/recommendations	POST	Tailored lifestyle and dietary clinical guidance based on individual risk tier.
/auth/login & /auth/register	POST	JWT Bearer security with bcrypt salted password encryption.
/history/{user_id}/export/csv	GET	One-click downloadable CSV patient trajectory for EHR integration.
/admin/models	GET	Active model registry and multi-algorithm benchmark comparison suite.

Clinical Disclaimer: *This document and the associated software are developed strictly for Clinical Decision Support (CDS) demonstration and portfolio evaluation. They do not constitute an autonomous medical device or diagnostic instrument. All clinical care must be supervised by a licensed medical practitioner.*